Compare and contrast Mitosis and Meiosis Worksheet

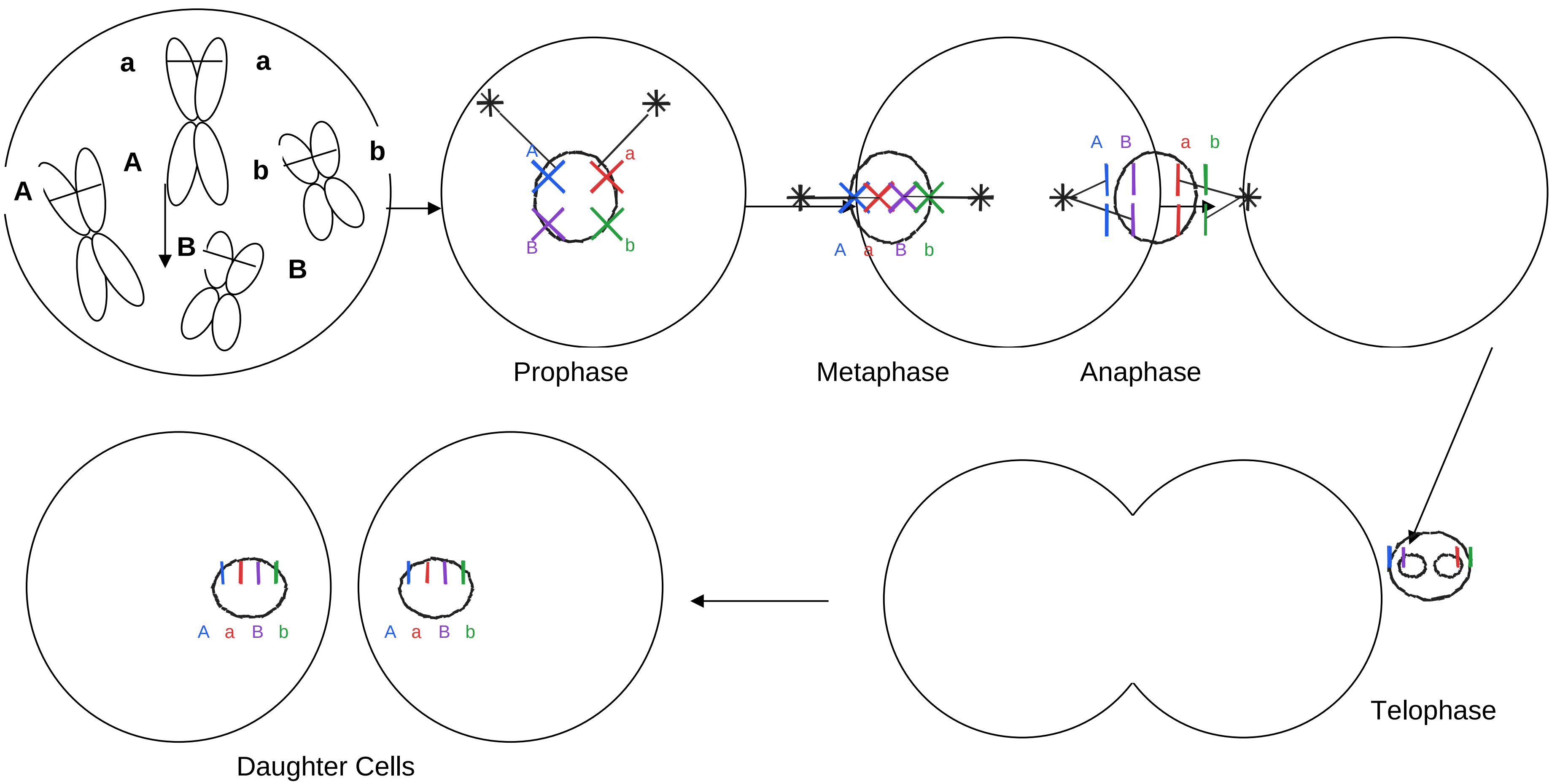
Part I <u>Mitosis</u>→ Use Diploid skin cells for Mitosis with 2n =4 (as shown in picture below)

Material:
colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

I. Draw the chromosomes in the cell as it undergoes **Mitosis**:

Remember homologus chromosomes line up on separate spindle fiber



Part II Meiosis

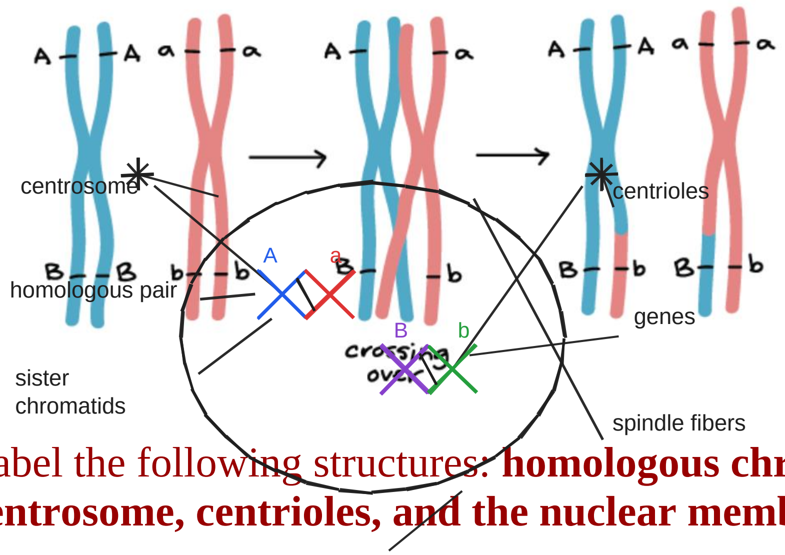
II. Draw the chromosomes in the cell as it undergoes **Meiosis (same parent cell):**

Use Use Diploid testical cells for Mitosis with 2n =4 (as shown in the picture below)
Remember Homologus chromosomes line up on separate spindle fibers

Material:
colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

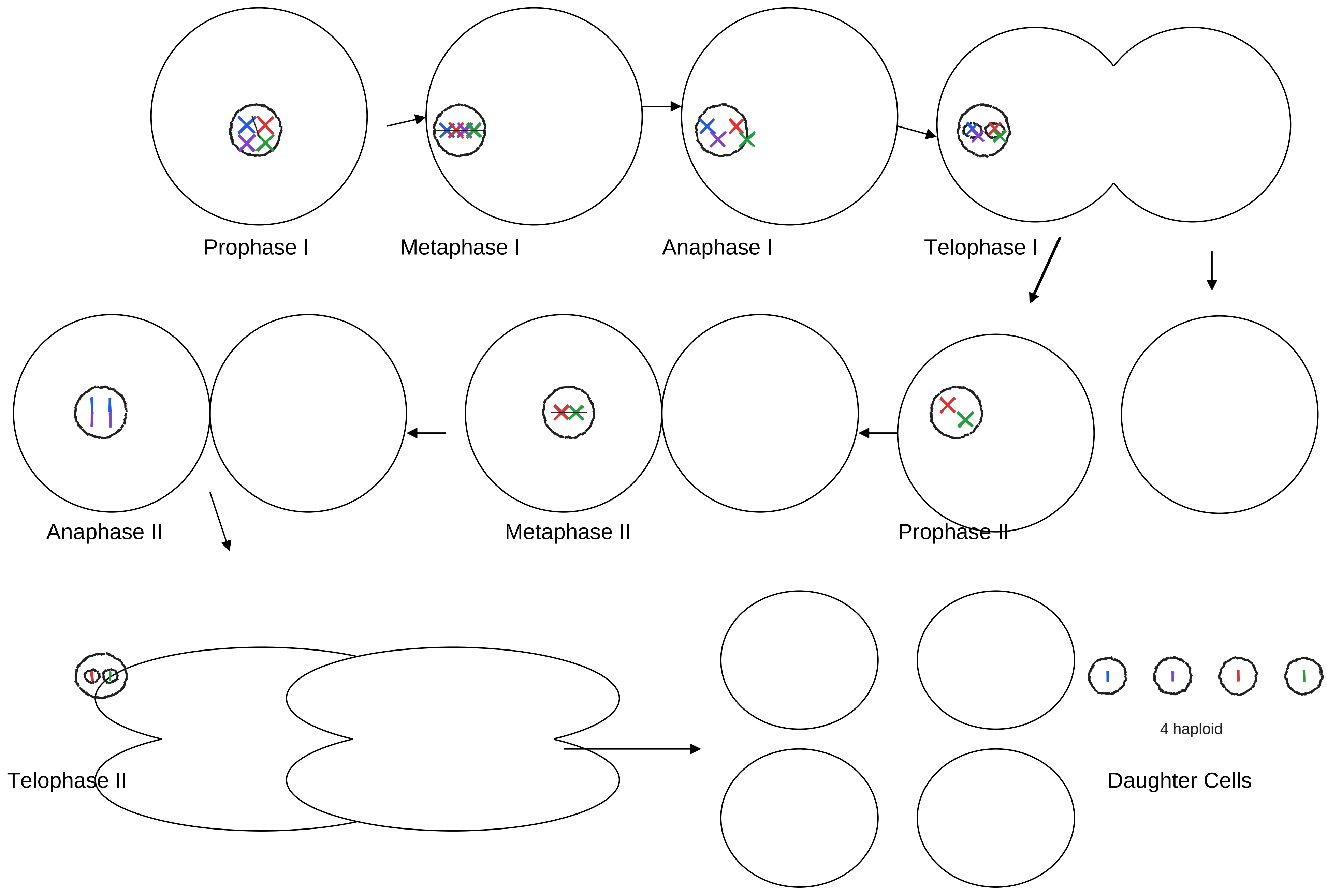
Show crossing over by switching the genes during Prophase I of meiosis



Note that chromosomes A, a are on one homologous pair of chromosomes in our exercise, and B, b are on a separate second homologous pair of chromosomes

Label the following structures: **homologous chromosomes, sister chromatids, genes, spindle fibers, centrosome, centrioles, and the nuclear membrane.**

You do **not** need to include all labels on a single cell—spread them across the diagram so each structure is clearly visible and easy to identify.


Prophase I
Metaphase I
Anaphase I
Telophase I
Anaphase II
Metaphase II
Prophase II
Telophase II
4 haploid
Daughter Cells

Part III: Complete the summary chart Comparing and Contrasting Mitosis and Meiosis

Cell Processes	**Mitosis**	**Meiosis**
Creates	body cells	sex cells (gametes)
Definition	1 division -> 2 identical 2n cells	2 divisions -> 4 different n cells
End Products	2 diploid cells	4 haploid cells
Steps	PMAT once	PMAT I + PMAT II
Type of Reproduction	asexual	sexual

Are they identical to the parent cell?	yes	no
When does cytokinesis occur?	once (after telophase)	twice (after I and II)
How many times does the parent cell divide?	1 time	2 times
What happens to the number of chromosomes at the end of each process? Are they in pairs or individual chromosomes?	stays 2n (46 in humans) in pairs	halves to n (23 in humans) individual
Why is each important?	growth + repair	variation + proper # for fertilization

How many chromosomes do human body cells and human sex cells have after they go through each process?	Mitosis: body cells = 46 Meiosis: sex cells = 23	

Part III

Explain **nondisjunction**—when chromosomes or chromatids fail to separate properly—and how it can lead to conditions such as **Down syndrome** and **Turner syndrome**.

Then explain how these chromosomal abnormalities result in different physical and developmental effects.

Nondisjunction = chromosomes do not separate right.
In meiosis I homologous chromosomes can fail to split;
meiosis II sister chromatids can fail to split.
This makes gametes with extra or missing chromosomes.
Down syndrome: trisomy 21 (47 total).
Turner syndrome: monosomy X (45, X).
These change gene dosage, so development can differ:
Down can include learning delays and heart risks;
Turner can include short stature and ovarian issues.